

PHISHING VILLAGE

11TH OCTOBER, 2025 @ cOcOn 2025

AI & Deepfake Phishing: How AI-powered content is reshaping social engineering

- Abhijeet

Who are we

- We're Phishing Village.
- Here to raise awareness on both offensive & defensive side of phishing.
- To help build tradecraft to conduct social engineering attacks as well as to prevent it.

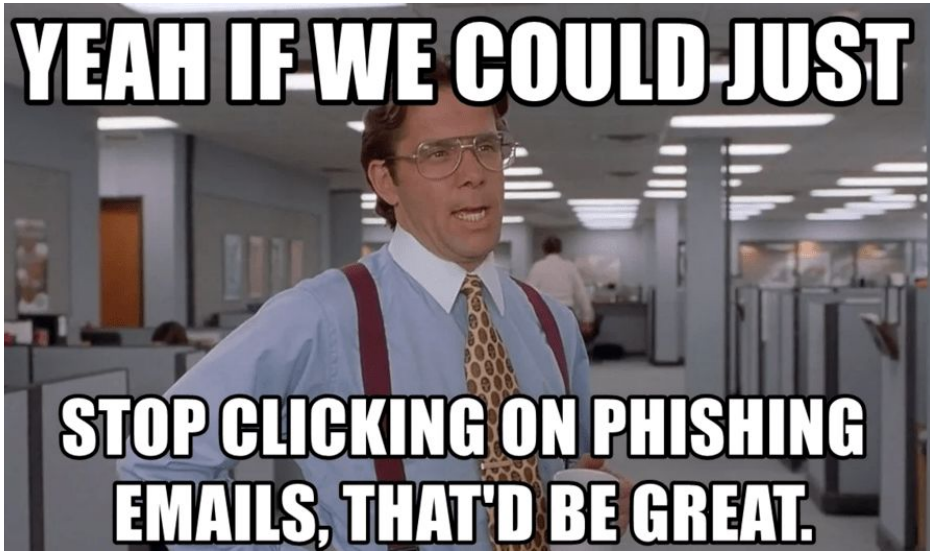


Whoami

- Security Researcher & Tinkerer
  - Research areas include Adversarial TTPs, Linux systems, and supply chain
- Current gig: Breaking things @CyberWarFare Labs
- Full-time human, part-time homecook

Phishing Types

- Traditional (Mass emails, fake sites, or malicious attachments)
- Spear (Targeted, personalized attacks)
- Smishing/Vishing (Phishing via SMS or phone calls)





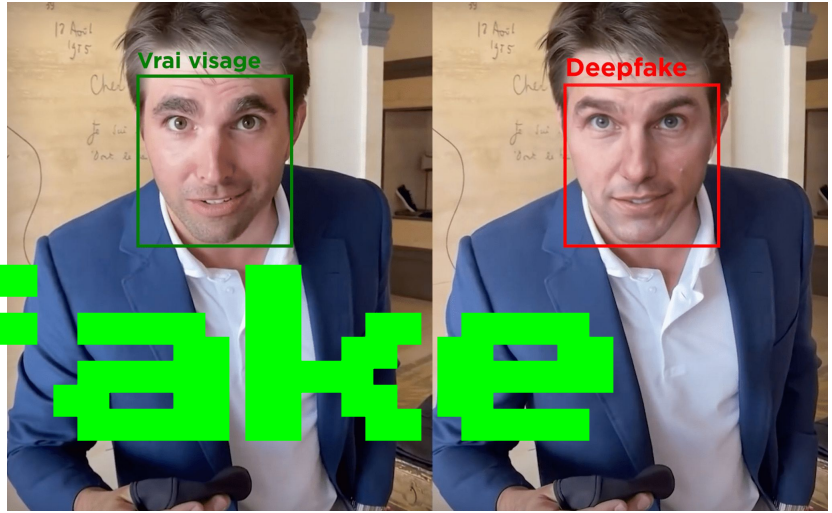
Now what the BLEEP is a deepfake?

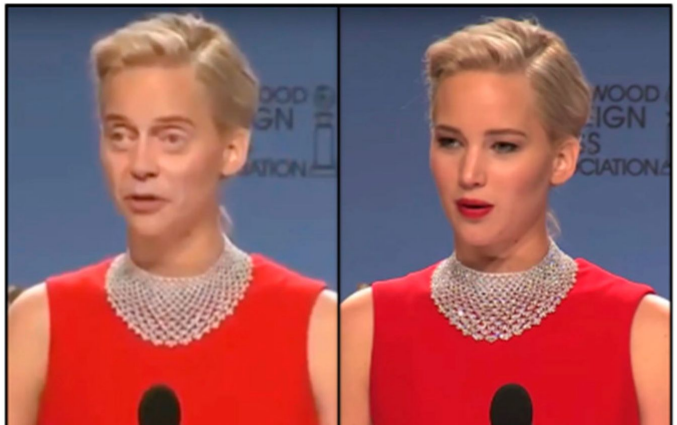






Deep fake







https://vimeo.com/1127653180#t=0

Where Do They Come From?


You get a LLM

Introduction to LLMs

- LLMs (Large Language Models) are subclass of artificial neural networks trained on massive datasets.
- Advanced LLMs comes under Generative Pre-trained Transformers (GPT).



Introduction to LLMs

- Generative AI → text, images, voice, video.
- Types of models :-
  - Text-based (ChatGPT, Llama, Claude)
  - Voice cloning (ElevenLabs, VALL-E)
  - Text-to-video (Pika Labs, Runway, Sora)

Deepfakes 101

- Comes from combination of Deep (Deep Learning) and Fake.
- Uses Artificial Intelligence (AI), specifically Machine Learning (ML) algorithms to generate realistic audio/video/images of humans.



Deepfake 101

- Deep Learning + Generative Models
- GANs (Generative Adversarial Networks)
- Diffusion Models (Stable Diffusion, Sora)
- Voice cloning → waveform + spectrogram models
- Video deepfakes → face swap + lip-sync + avatar gen

Deepfakes 101

1990s

Computer Generated Imagery (CGI)

Researchers started using CGI to generate human images.

2014

Generative Adversarial Networks (GANs)

Ian Goodfellow developed the concept where two neural networks compete in a profit-loss game.

2017

Deepfake

The keyword “Deepfake” was coined by a Reddit user.

2018

Deepfake booming

Deepfakes started popping into public domain.

2021

DALL-E release

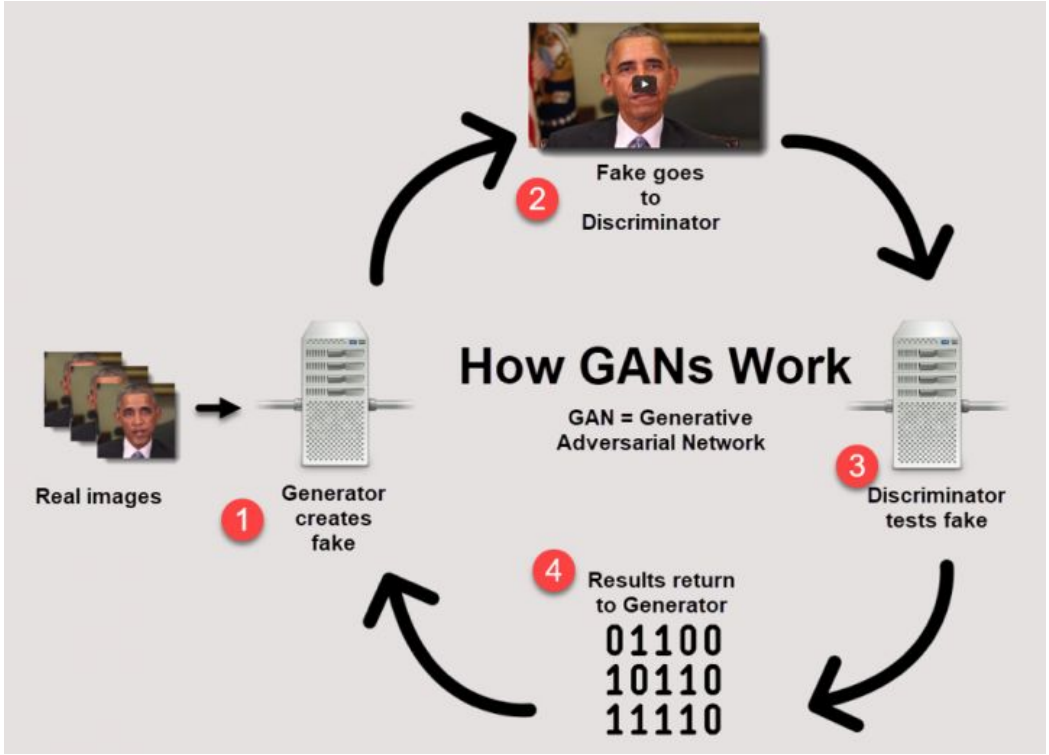
Text-to-image model debut in public domain.

2022-2025

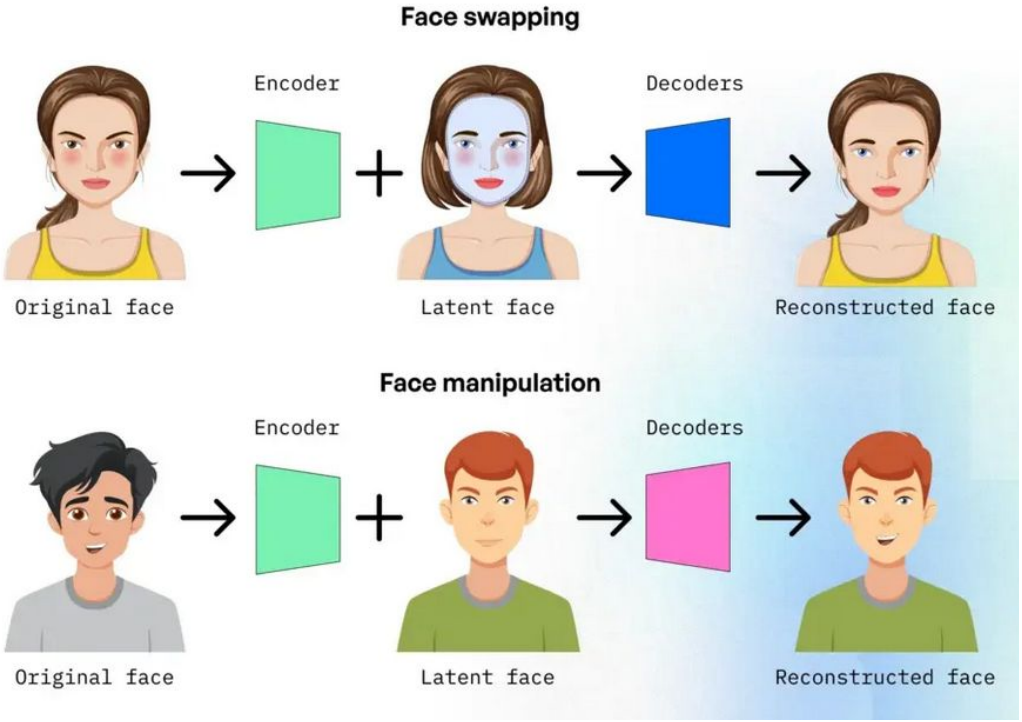
Chaos

ChatGPT & other GenAI + LLM tools released, easing the creation of deepfakes for both technical & non-technical folks.

Generative Adversarial Networks (GANs)



Deepfakes Types





Ever used Instagram or Snapchat face filters?

Why AI Makes Social Engineering Better?

- Generates personalized, fluent content instantly
- Removes language/grammar red flags
- Voice & face cloning enables synthetic trust
- Scales targeted phishing



Case Study 1: North Korean Fake IT Worker

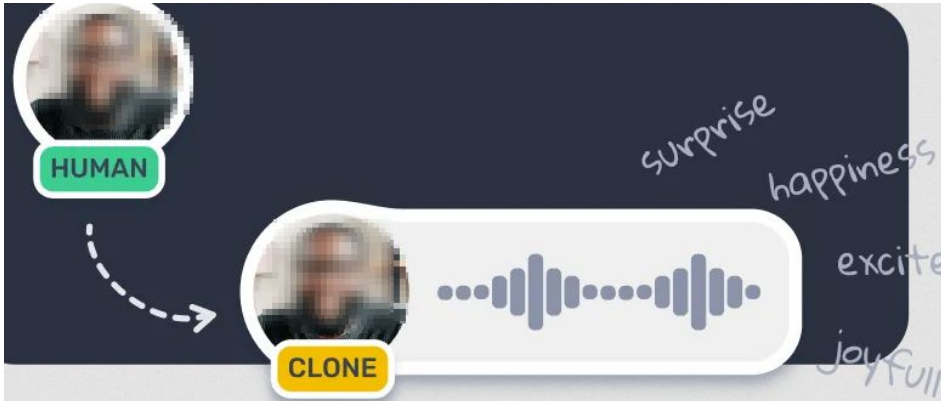
- Threat Actor used stolen ID.
- AI-enhanced picture.
- Cleared video conferencing rounds.
- Got caught by SOC while loading malware.





Case Study 2: Voice Cloning CFO Scam

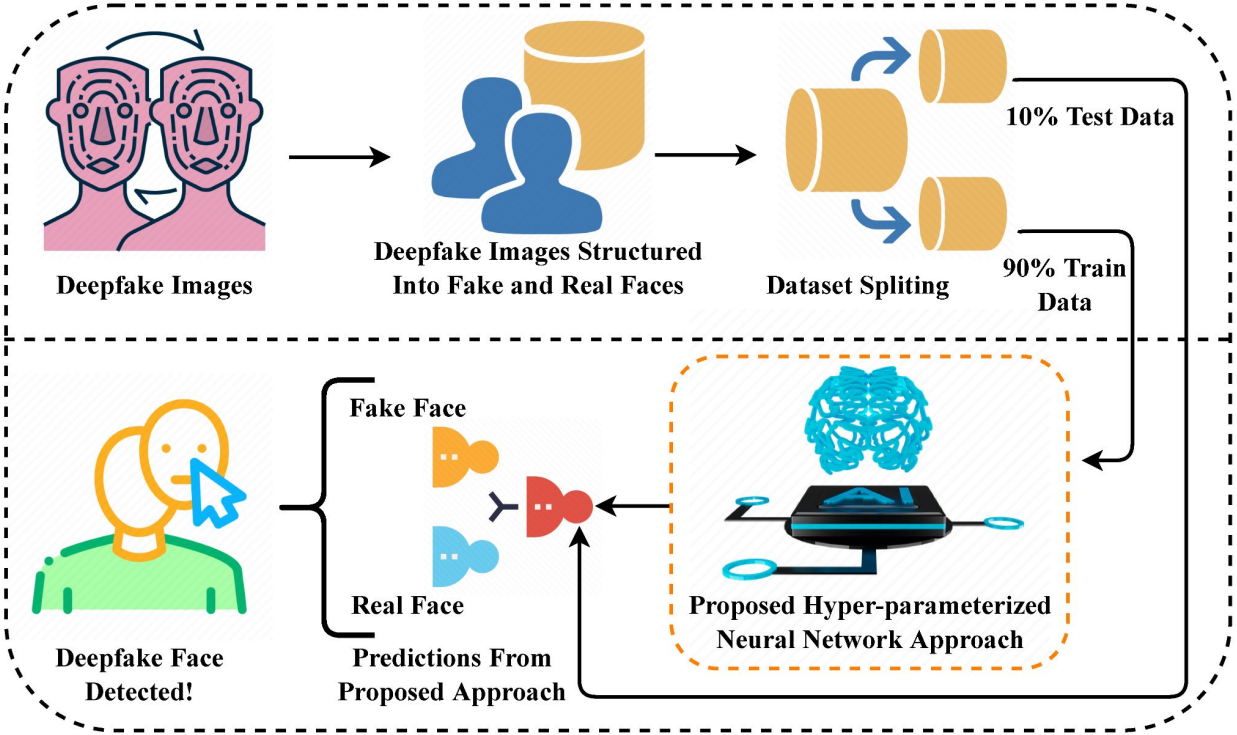
- TA obtained voice samples.
- Created cloned voice.
- Hopped on a Zoom call with background noise & CEO's voice.
- Tricked finance team into transferring money.



Generate Your Own Deepfakes (GYOD)

- **Voice Cloning:-**
  - ElevenLabs
  - VALL-E
  - Respeecher
- **Video Deepfakes:-**
  - DeepFaceLab
  - DeepFaceLive
  - FaceSwap
  - Synthesia

Deepfake Detections





Takeaways

- Phishing has moved from text → voice → video
- AI removes traditional red flags
- Real-world deep fake phishing attacks have already caused millions in damages
- Defense requires multi-factor verification + awareness
- Slides Link: https://github.com/wand3rlust/My-Presentations

Thank You